

DIGITAL FORENSICS

FINAL INVESTIGATION REPORT

Microsoft Office 2013 Password Recovery Analysis

Case Reference No.	DFIR-2026-003
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1. Overview

This examination documents the analysis of a password-protected Microsoft Office 2013 Word document and the subsequent recovery of the password through offline password-cracking techniques. The objective of the examination was to determine the encryption type used by the document, extract the Office password hash, perform controlled password-recovery attempts, and verify successful decryption of the document.

The workflow included identifying the Office 2013 encryption format, extracting the structured \$office\$ hash using office2john, preparing the hash for cracking tools, confirming the appropriate Hashcat mode, and conducting dictionary-based attacks using Hashcat and John the Ripper. Successful recovery of the password was confirmed by opening the encrypted document and validating access to its contents.

This report records the steps performed, commands executed, tools used, and the resulting findings from the examination.

2. Methodological Note

This examination was conducted during my early practical training period in digital forensics. At the time of the analysis, a cryptographic integrity hash (for example, SHA-256) of the original Microsoft Word document was not calculated prior to examination.

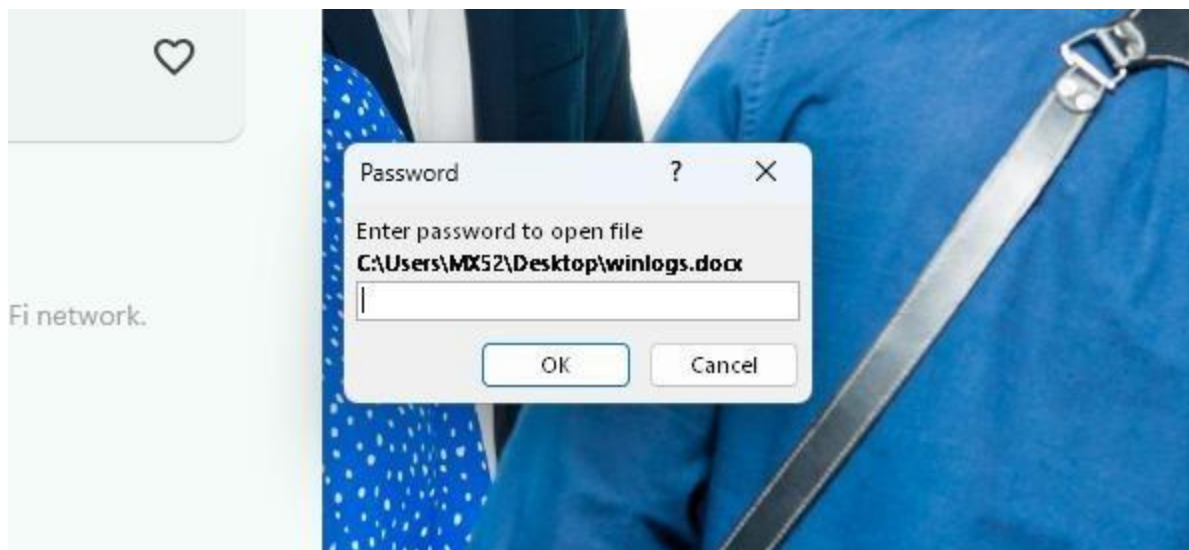
The omission occurred because I was still developing my forensic workflow and was not yet aware of the importance of documenting evidence integrity before analysis. My current forensic practice includes hashing evidence files before examination and recording the resulting hash values as part of the chain of custody and integrity verification process.

No retrospective hash values have been added to this report.

3. Encrypted Document Prompt (Before Cracking)

This image shows the password-protected Microsoft Word document upon opening. The prompt confirms that the file is encrypted using Microsoft Office 2013 document encryption. At this stage, access to the file contents was restricted pending successful password recovery.

The presence of encryption indicates that the document uses a key derivation function with SHA-256 and 100,000 iterations, as later confirmed during hash extraction.



4. Hash Extraction

The encrypted document was processed using office2john to extract the password hash in a format suitable for offline cracking. The tool generated a structured hash beginning with:

\$office\$*2013*100000*256*16*...

This signature confirms that the document uses Microsoft Office 2013 encryption. The extracted hash includes:

- Version identifier (2013)
- Iteration count (100,000)
- Key size (256-bit)
- Salt value
- Encrypted verifier and hash data

The full structured hash was saved to a text file, ensuring the \$office\$ prefix was preserved to maintain compatibility with cracking tools.

```
kali@kali: ~/Desktop
(kali@kali)-[~/Desktop]
$ office2john winlogs.docx
winlogs.docx:$office$*2013*100000*256*16*70ed6ecef478ce6370382bff17ee4766*338f42d9d88ce879f1203449ce1c0527*f70f69dc218520145bb8944b05b68a2281f9fd6e784348be94da1d6b2efe3ffe
```

5. Preparing the Hash File

After Hash Extraction – Preparing the Hash File

After extracting the hash with office2john, I saved the full \$office\$ structured hash into a text file called **officehash.txt**.

A terminal window with a dark background and light blue text. The prompt is (kali@kali)-[~]. The command entered is echo '\$office\$*2013*100000*256*16*70ed6ecef478ce6370382bff17ee4766*338f42d9d88ce879f1203449ce1c0527*f70f69dc218520145bb8944b05b68a2281f9fd6e784348be94da1d6b2efe3ffe' > officehash.txt. The output is officehash.txt.

```
(kali@kali)-[~]  
$ echo '$office$*2013*100000*256*16*70ed6ecef478ce6370382bff17ee4766*338f42d9d88ce879f1203449ce1c0527*f70f69dc218520145bb8944b05b68a2281f9fd6e784348be94da1d6b2efe3ffe' > officehash.txt
```

It was important to keep the entire line exactly as generated — including the \$office\$ prefix.

Without it, cracking tools like Hashcat would not recognize the format.

I used:

```
echo '$office$*2013*100000*256*16*...' > officehash.txt
```

6. Verifying the Correct Hashcat Mode

Before starting the attack, I confirmed the correct Hashcat mode for the extracted hash.

I ran:

```
hashcat --help | grep -i office
```

The output shows different Microsoft Office versions and their corresponding mode numbers.

Since the extracted hash contained *2013*, I selected:

Mode 9600 – MS Office 2013

This step is important. Using the wrong mode (like raw SHA-256) would fail, even if the password is correct.

```
(kali@kali)-[~/Desktop]
$ hashcat --help | grep -i office
 9400 | MS Office 2007 | Document
 9500 | MS Office 2010 | Document
 9600 | MS Office 2013 | Document
25300 | MS Office 2016 - SheetProtection | Document
 9700 | MS Office <= 2003 $0/$1, MD5 + RC4 | Document
 9710 | MS Office <= 2003 $0/$1, MD5 + RC4, collider #1 | Document
 9720 | MS Office <= 2003 $0/$1, MD5 + RC4, collider #2 | Document
 9810 | MS Office <= 2003 $3, SHA1 + RC4, collider #1 | Document
 9820 | MS Office <= 2003 $3, SHA1 + RC4, collider #2 | Document
 9800 | MS Office <= 2003 $3/$4, SHA1 + RC4 | Document
```

7. Cracking the Hash with Hashcat

The extracted Office 2013 hash was attacked using Hashcat in mode 9600, which corresponds specifically to Microsoft Office 2013 encryption.

Command used:

hashcat -m 9600 officehash.txt /usr/share/wordlists/rockyou.txt

```
(kali@kali)-[~/Desktop]
$ hashcat -m 9600 officehash.txt /usr/share/wordlists/rockyou.txt

hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, LLVM 17.0.6, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]

$office$*2013*100000*256*16*70ed6ecef478ce6370382bff17ee4766*338f42d9d88ce879f1203449ce1c0527*f70f69dc218520145b
b8944b05b68a2281f9fd6e784348be94da1d6b2efe3ffe:keyboard

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 9600 (MS Office 2013)
Hash.Target.....: $office$*2013*100000*256*16*70ed6ecef478ce6370382bf...fe3ffe
Time.Started....: Sun Feb 15 20:00:57 2026 (1 min, 13 secs)
Time.Estimated...: Sun Feb 15 20:02:10 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 42 H/s (3.30ms) @ Accel:1024 Loops:16 Thr:1 Vec:4
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 3072/14344385 (0.02%)
Rejected.....: 0/3072 (0.00%)
Restore.Point....: 2048/14344385 (0.01%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: slimshady -> dangerous
Hardware.Mon.#1...: Util: 72%

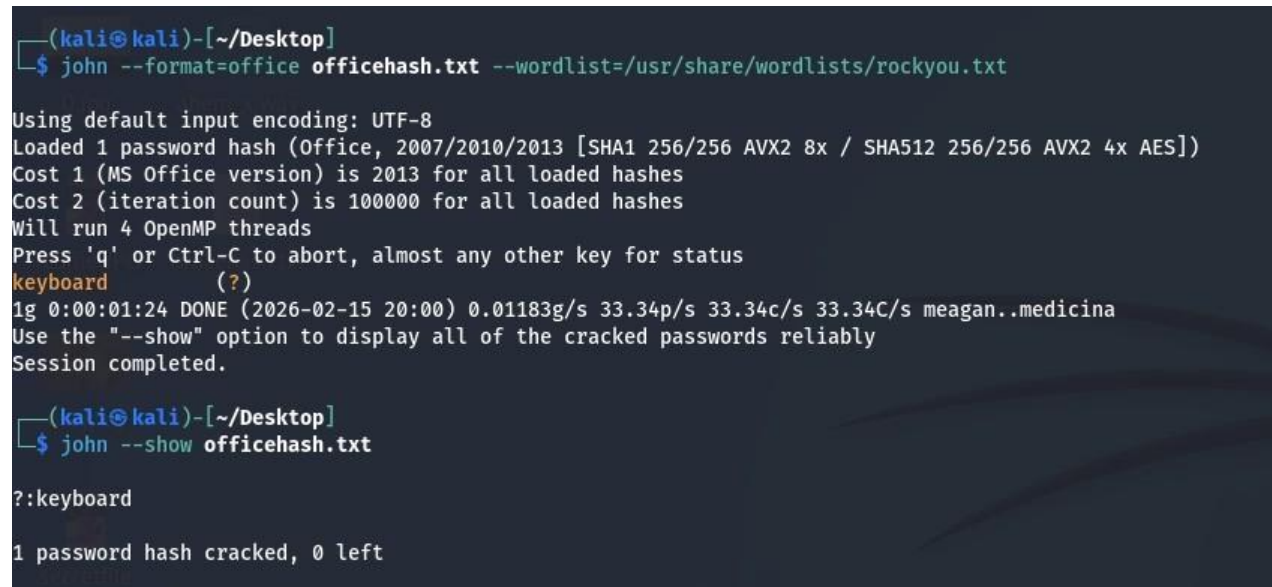
Started: Sun Feb 15 19:58:31 2026
Stopped: Sun Feb 15 20:02:13 2026
```

8. Cracking the Hash with John the Ripper

The extracted Office 2013 hash was attacked using JohnTheRipper, which corresponds specifically to Microsoft Office 2013 encryption.

Command used:

```
john --format=office officehash.txt --wordlist=/usr/share/wordlists/rockyou.txt
```

A screenshot of a terminal window with a dark background. The prompt is (kali@kali)-[~/Desktop]. The command john --format=office officehash.txt --wordlist=/usr/share/wordlists/rockyou.txt is entered. The output shows: Using default input encoding: UTF-8, Loaded 1 password hash (Office, 2007/2010/2013 [SHA1 256/256 AVX2 8x / SHA512 256/256 AVX2 4x AES]), Cost 1 (MS Office version) is 2013 for all loaded hashes, Cost 2 (iteration count) is 100000 for all loaded hashes, Will run 4 OpenMP threads, Press 'q' or Ctrl-C to abort, almost any other key for status, keyboard (?), 1g 0:00:01:24 DONE (2026-02-15 20:00) 0.01183g/s 33.34p/s 33.34c/s 33.34C/s meagan..medicina, Use the "--show" option to display all of the cracked passwords reliably, Session completed. The prompt is (kali@kali)-[~/Desktop]. The command john --show officehash.txt is entered. The output shows: keyboard, 1 password hash cracked, 0 left.

```
(kali@kali)-[~/Desktop]
$ john --format=office officehash.txt --wordlist=/usr/share/wordlists/rockyou.txt

Using default input encoding: UTF-8
Loaded 1 password hash (Office, 2007/2010/2013 [SHA1 256/256 AVX2 8x / SHA512 256/256 AVX2 4x AES])
Cost 1 (MS Office version) is 2013 for all loaded hashes
Cost 2 (iteration count) is 100000 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
keyboard (?)
1g 0:00:01:24 DONE (2026-02-15 20:00) 0.01183g/s 33.34p/s 33.34c/s 33.34C/s meagan..medicina
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

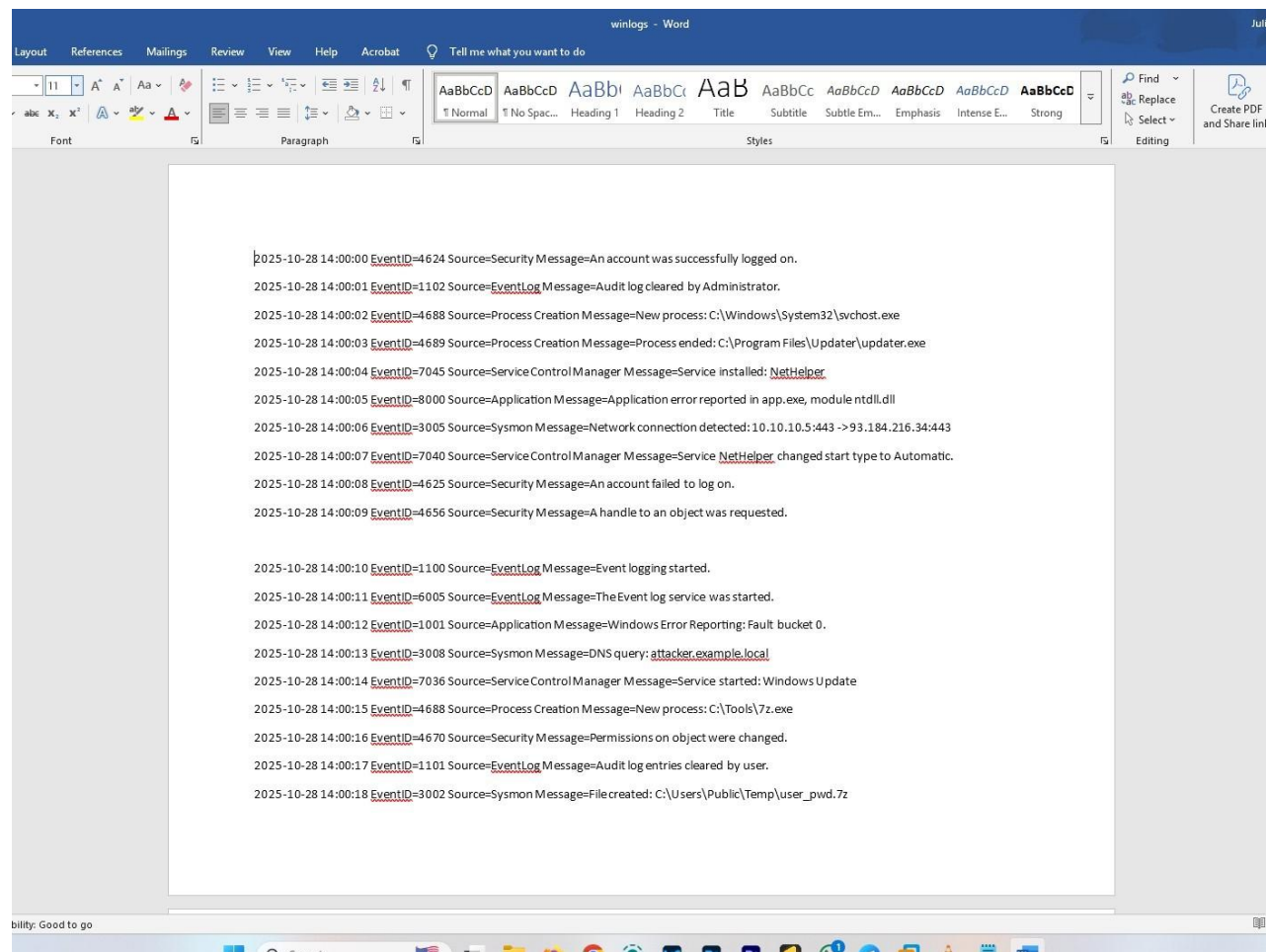
(kali@kali)-[~/Desktop]
$ john --show officehash.txt

?:keyboard

1 password hash cracked, 0 left
```

9. Successful Document Decryption

Following password recovery, the document was opened successfully using the cracked password. This confirms that the recovered credential correctly derived the encryption key required to decrypt the document contents.



10. Tools Used

Microsoft Word, office2john, Hashcat, John the Ripper, rockyou.txt, Bash (Linux command line)

Since the extracted hash contained *2013*, I selected:

11. Validation of Findings

The successful decryption validates:

- Correct hash extraction
- Proper identification of encryption type
- Accurate selection of Hashcat mode (9600)
- Effective wordlist-based attack